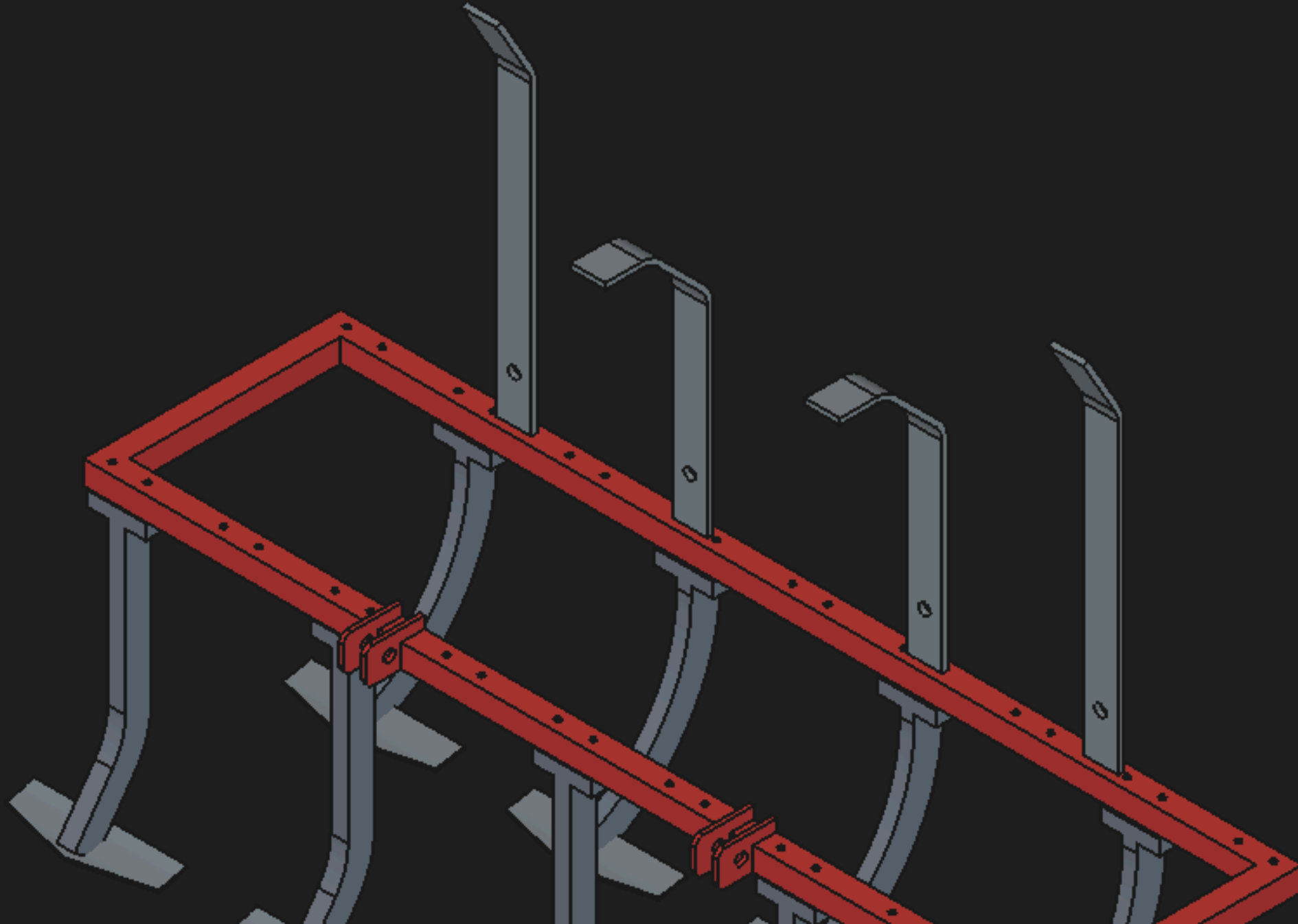
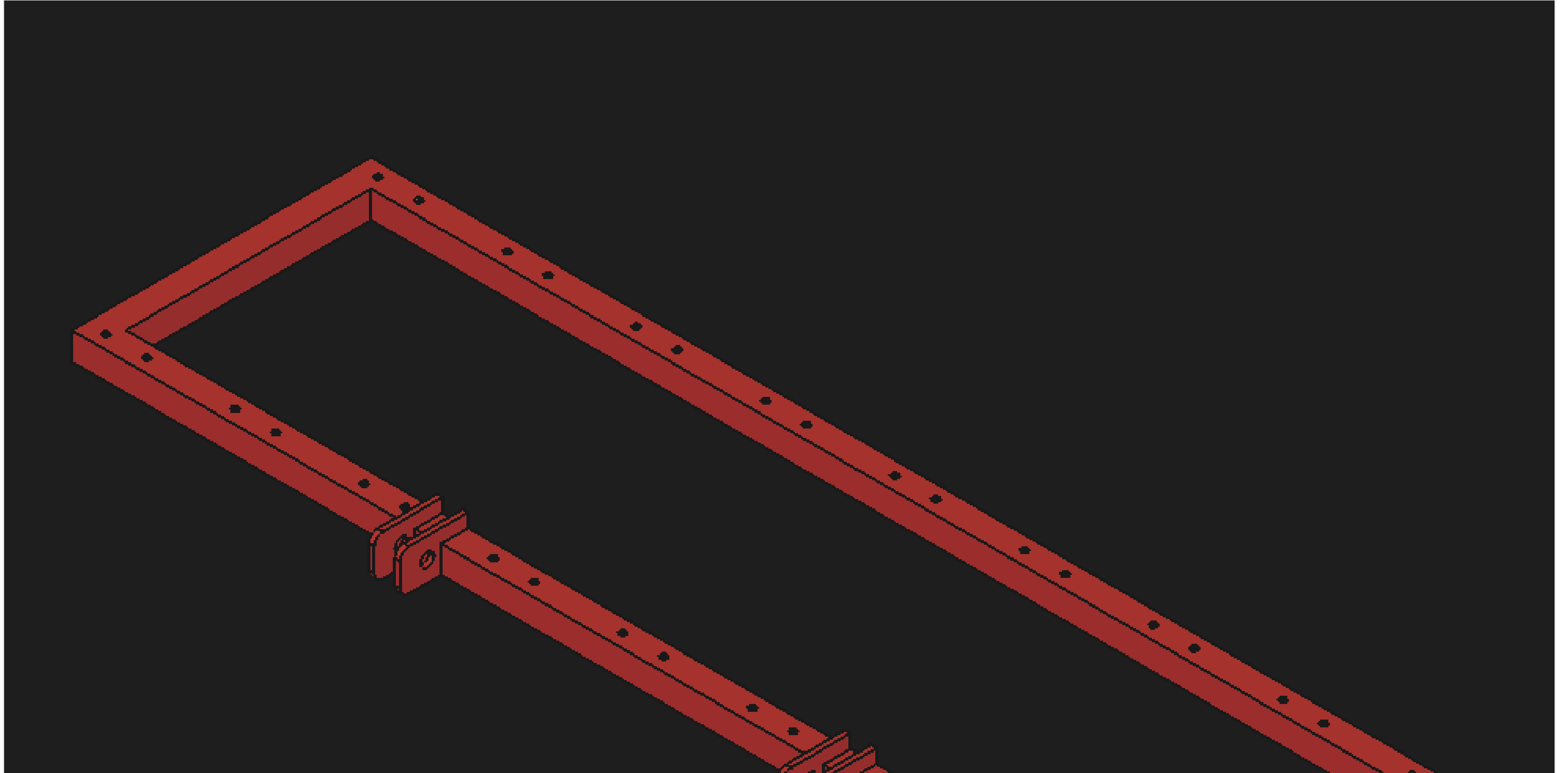


Summary and Exercises



SEVEN PARTS

What you built



Ten things worth remembering

- **Fully constrain every sketch.** No exceptions.
- **Put sketches on origin planes,** with offsets. Not on faces.
- **Use endpoint tangency** where a line meets a curve, and nothing else at that corner.
- **Prefer distances to angles.** An angle has two answers.
- **Read the message in the Sketcher panel.** It tells you what is wrong.
- **A pocket cuts from its sketch plane onward.** Put the plane where the cut starts.
- **You cannot pattern a pattern.** Do the symmetry in the sketch.
- **Fillets and chamfers go last,** and keep them few.
- **Put each part's origin on the face that matters** — the one that bolts to something else.
- **Every dimension in a spreadsheet,** named, with its unit.

Everything we assumed

WE DECIDED		BECAUSE THE SOURCE
1	Frame holes are pairs, 3 in apart	said “one hole and another”
2	Tube is 2 by 2 in, 3/16 wall	gave only the wall
3	Bracket is 5 in long, 3 in tall	never dimensioned it
4	Tine 01 rakes 35 degrees	did not say
5	Tine 01 bends on R6	did not say
6	25 in is total, 15 in is the straight part	gave both, unrelated
WE DECIDED		BECAUSE THE SOURCE
7	Tyne bolt holes go through	said blind 0.4 in
8	Clamps are braces for a mast	never mated them
9	Clamp 02 tip is 6 in long	gave the angle only
10	Blade radius is on the centreline	used Flex conventions
11	Blades have no bolt holes	never mentioned them
12	Two corner fillets, not a full round	asked for a full round

All twelve are spreadsheet numbers. Disagreeing with any of them costs one cell edit.

Exercises: get comfortable

PARAMETER CHANGES

- 1 **Eleven tynes.** Set the count to 11 and the frame to 101 inches. Report the new weight.
- 2 **A ten-foot machine.** Decide the spacing first, then work out the count and the length.
- 3 **Re-rake Tine 01** to 25 and then 45 degrees. Which would you specify for hard ground, and why?
- 4 **Add blade bolt holes.** You decide the spacing — and justify it.

MODELLING JOBS

- 1 **Model a bolt:** half inch by four inches, hex head. Then place eighteen of them.
- 2 **Draw the frame** on a TechDraw sheet, with a section through a tool bar.
- 3 **Design a third tyne** — a spring tine that uses the same 5 inch head and 3 inch bolts.
- 4 **Sharpen the blade.** Add a ground edge underneath the two cutting edges.

Exercises: harder

TAKE IT FURTHER

- 1 **Build a proper hitch.** The machine as modelled has no top link point. Design an A-frame mast that ties into the brackets and the rear bar.
- 2 **One master parameter table.** Move the bolt size and spacing into a single file that the frame and both tynes read from. Then change the bolt to 5/8 inch.
- 3 **Assemble it with joints** instead of placements. Report anywhere the parts did *not* fit.

THE BIG ONE

- 1 **Analyse a tyne.** Take Tine 01 into the **FEM** workbench. Fix the head, push 150 pounds sideways at the point, and solve for stress. Is the section right, over-built, or marginal?

Exercise 12 is where this stops being drawing and starts being engineering. A model built the way this course built it — right material, right geometry, right weight — is ready for it.

How your work is judged

